

Learning Plan - MySQL, C#, .NET, Next.js

Day 1

- MySQL: Introduction, Installation & Setup
- Basic Commands: CREATE, SELECT, INSERT, UPDATE, DELETE
- Data Types & Constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK)

Day 2

- MySQL: WHERE, ORDER BY, LIMIT, DISTINCT
- Joins (INNER, LEFT, RIGHT, FULL)
- GROUP BY, HAVING, Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

Day 3

- MySQL: Functions (String, Date, Numeric)
- Views (Create, Update, Drop) & Types of Views
- Normalization & its Types (1NF, 2NF, 3NF, BCNF)
- Common Table Expressions (CTEs)
- Transactions & ACID Properties

Day 4

- C#: Introduction & Syntax
- Variables, Data Types, Operators
- Control Statements (if, switch, loops)
- Methods & Parameters

Day 5

- C#: Object-Oriented Programming (Classes, Objects, Constructors)
- Encapsulation, Inheritance, Polymorphism, Abstraction
- Interfaces & Abstract Classes

Day 6

- C#: Collections (List, Dictionary, Queue, Stack)
- Exception Handling

- File Handling

Day 7

- .NET: Introduction, CLR, Framework
- Creating a basic .NET Console App
- Understanding NuGet Packages

Day 8

- .NET: Web Development Basics
- ASP.NET Core MVC
- Routing & Controllers

Day 9

- Next.js: Introduction & Installation
- Pages & Routing
- Static & Server-Side Rendering

Day 10

- Next.js: API Routes
- Fetching Data (getStaticProps, getServerSideProps)
- Basic Deployment